Leicht·Facile

Easy-language drafts of Swiss official texts with Apertus, with every fact value checked by code.

TEAM
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REPOSITORY
github.com/bobrovsky420/leicht-facile

DEMO
youtu.be/V_R4AABLriI

MATURITY
deployable prototype for German and French drafts of public texts and letters; proof of concept for court and KESB decisions; image input explored, not built.

100 %	97 %	0 %	4.2
fact coverage	**hard facts kept**	**silent losses**	**rule errors**
must-keep facts found	the source's countable values	values lost without a trace	per 100 blocks
Strong single prompt: 83 %	Strong single prompt: 85 %	Strong single prompt: 15 %	Strong single prompt: 15.0

Our pipeline (P2) against the strong single prompt (P0-ZH complete) on the same model and the same held-out pairs; §5 has the setup and every row.

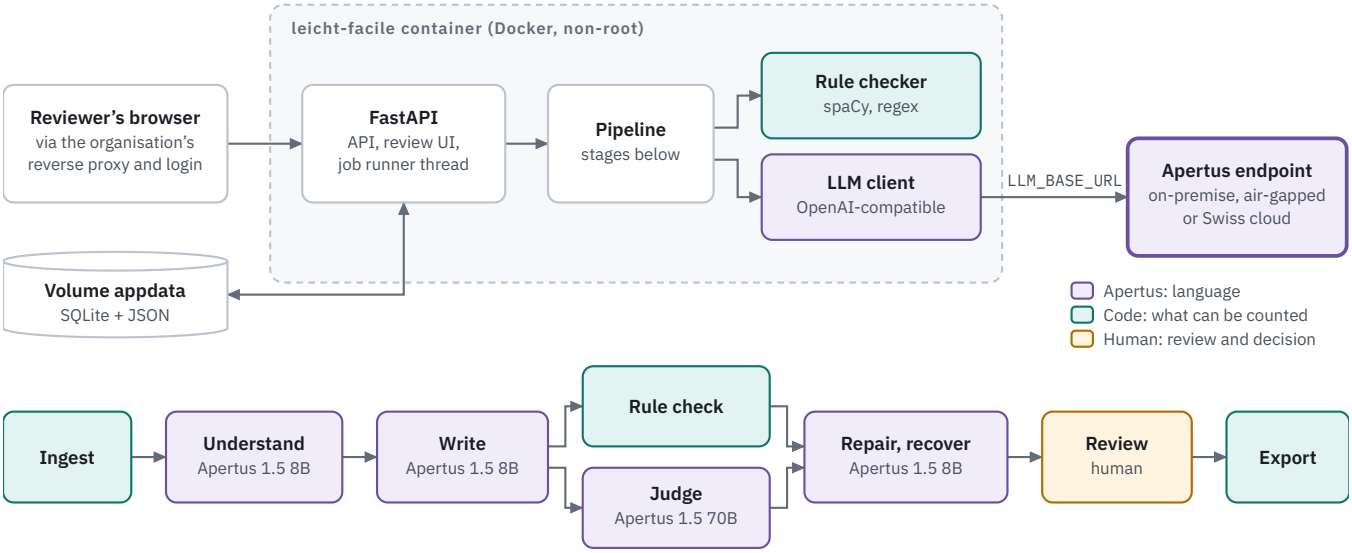
1. Summary

Problem. About 1.25 million adults in Switzerland aged 16 to 65 (22%) have low reading skills (Swiss Federal Statistical Office, PIAAC 2022/23); people with intellectual disabilities, language learners and people with dementia need easy language too. Disability law (BehiG and the UN disability convention) asks authorities for accessible information, and cantons have begun to publish in Leichte Sprache, FALC or lingua facile; each version is written by hand by a few trained writers, who must check every deadline, amount and phone number against the original.

Approach. Leicht·Facile is a drafting assistant for the people who write official texts: trained writers, and the case workers, clerks and physicians who write decisions and letters themselves. One rule: **Apertus does everything that needs language understanding; plain code does everything that can be counted.** Apertus 1.5 turns the document into a fact sheet, writes a draft with provenance for every sentence, judges meaning with a second, larger model, repairs only the flagged sentences and asks again for each value still missing. Code checks the published easy-language rules (45 checks for German, 41 for French), including the exact presence of every fact value. A person reviews source and draft side by side and exports with an audit trail. One container plus an OpenAI-compatible Apertus endpoint runs on-premise, air-gapped or on a Swiss cloud, as letters under official, medical or tax secrecy require; for those, only Apertus fits (§3). A single prompt ("rewrite this in Leichte Sprache"), with no fact contract, provenance or independent check, is our baseline P0; the full pipeline is P2.

Headline result. Same model, a strong single prompt (defined in §5) against our pipeline, on 20 held-out German pairs on the CSCS endpoint (final run, 11 October 2026): the pipeline finds 100 percent of the must-keep facts against 83, loses none without a trace against 15 percent, and has 4.2 against 15.0 rule errors per 100 blocks; invented values are rare on both sides (0.1 against 0.0). It keeps 97 percent of the source's countable values (dates, amounts, phone numbers, addresses) against 85 for that prompt in its complete mode and 67 for the official easy-language versions, which drop values on purpose, and lists every one it loses; the prompt in its condensed default loses 33 percent without a trace.

2. Architecture



Stage	Done by	Output
Ingest	Code: trafilatura, pdfplumber, spaCy	`SourceDocument`: segments with stable ids (`s0001`)
Understand	Apertus 8B, thinking	`FactSheet`: the reader's core questions, typed facts with normalised values and source ids; code adds every date, amount, contact, address, period or identifier the model missed
Write	Apertus 8B	`Draft`: blocks with the facts and source segments they carry, omissions with reasons, glossary
Rule check	Code: spaCy parse, regex, word frequencies	Findings with rule id, severity, evidence, suggestion
Judge	Apertus 70B, thinking	Each fact present, missing or distorted; unsupported blocks; meaning findings. The judge never rewrites
Repair	Apertus 8B	New text for flagged blocks only; up to 3 rounds; every version kept
Recover	Apertus 8B	One sentence for each must-keep value still missing (at most 10), checked and judged like a round
Review, export	Human, then code	Per-block decisions; Markdown, HTML and a JSON audit bundle

Key decisions (track_2b/docs/architecture.md): countable rules in code, meaning in the model (exact, free, reproducible, no self-grading; track_2b/docs/rules.md). Every must-keep value must appear in an accepted form (`31. März 2026`, `31.3.2026`), so a deadline that became 30 March is caught; every value the source does not contain is flagged. Typed artifacts in SQLite let a failed job resume and the UI and the evaluation show every intermediate result. One service: FastAPI, a server-rendered UI, one worker thread by default; no Node build, no CDN.

Target architecture (mandatory)

Leicht·Facile is deployable in all three architectures. Its only runtime dependency is the Apertus endpoint in `LLM_BASE_URL`; changing the deployment means changing that variable.

Architecture	How it is met
a) On-premise	`docker compose --profile local-llm up` adds a vLLM service from the official image `ghcr.io/swiss-ai/vllm_apertus_1.5_release` serving Apertus 1.5 8B on one GPU (run on an L4 24 GB, below); the app points at `http://vllm:8000/v1`. A llama.cpp profile serves a 4-bit 8B on an 8 GB laptop GPU.
b) Air-gapped	As (a), with images and weights pulled in advance. Python packages, spaCy models and UI assets are baked into the image; no telemetry, no CDN, no installs at runtime.
c) Sovereign Swiss cloud	`LLM_BASE_URL` points at a Swiss-operated Apertus endpoint; the hackathon demo runs against the CSCS inference API in Lugano.

What the running system contacts:

Service	When	What it receives
Apertus endpoint (`LLM_BASE_URL`)	Every model call; its model list once per process	Document text, fact sheet and drafts, inside prompts
The server of a URL the user enters	Only for URL input	A request for that page; no document data

Nothing else: no telemetry, CDN, outside fonts or scripts, update check or login provider; on-premise or air-gapped, no data leaves the organisation at runtime. Build time: PyPI (`uv.lock`), spaCy wheels from GitHub, images from Docker Hub and GHCR. The container runs as non-root, logs no document text above `DEBUG` and listens on localhost; published further, it insists on a shared token or the reverse proxy's user header (the organisation's login). Licences:

Apertus 1.5 Apache 2.0 with an acceptable use policy (accepted once for the gated download), code Apache 2.0, spaCy models MIT (German) and LGPL-LR (French), fonts OFL. Any OpenAI-compatible server works (used: CSCS, Public AI, llama.cpp; vLLM on-premise): no single provider is needed.

Hardware for (a) and (b), measured on one NVIDIA L4 24 GB (`track_2b/eval/hardware/`): the 8B in bf16 takes 17.2 GiB, and with the profile's FP8 KV cache one 32k request fits. It generates 16 tokens per second, close to the limit of the card's memory bandwidth. A laptop RTX 4060 8 GB (llama.cpp, 4-bit) gave about 37 tokens per second.

3. Use of Apertus

- **Models:** `swiss-ai/Apertus-v1.5-8B` writes (Understand, Write, Repair, Recover, baseline) and `swiss-ai/Apertus-v1.5-70B` judges, set by `LLM_NAME` and `LLM_JUDGE_NAME` (without it, the judge is the 70B when the endpoint serves it). `LLM_UNDERSTAND_NAME` gives the Understand stage alone another model.
- **How it is used:** inference, as a structured-output agent pipeline without tool calling (the thinking variants are served with tools disabled). No fine-tuning.
- **Where it runs, and why only Apertus:** the CSCS inference API for the hackathon; the same client works unchanged with vLLM, llama.cpp or a Swiss cloud endpoint. Letters that name a person fall under official, medical or tax secrecy and may not go to a foreign cloud, so the organisation must run the model itself and account for what it learned. Only Apertus combines open weights, open training data (website opt-outs honoured), Swiss public origin and German, French and Italian (track_2b/docs/use-cases.md §2).
- **Why these sizes:** the 8B is enough to read and write: it finds as many must-keep facts as the 70B running every stage (100 against 99 percent) at half the price per page (§5), and runs on one 24 GB GPU (§2), where the 70B needs four. The 70B judges, because meaning needs a second, larger model than the one that wrote the draft. The 70B writes cleaner (0.6 against 4.2 rule errors per 100 blocks) but keeps fewer countable values (92 against 97 percent); the 70B reading for the 8B halves the rule errors (1.9) at a quarter more than the 8B's price, for an organisation that runs the 70B anyway.

Stage	Model	Thinking	Temperature	max_tokens
Understand	8B	on	0.1	12000
Write	8B	off	0.3	8000
Judge	70B	on	0.0	12000
Repair	8B	off	0.2	6000
Recover	8B	off	0.2	1500
Baseline P0	8B	off	0.3	6000

Prompts and parsing (track_2b/docs/prompts.md), in the document's language, each with one filled JSON example instead of the schema. Sentences and blocks carry ids (`[s0004]`, `[b007]`): the model cites, code verifies; the judge's unsupported verdicts must quote words that stand in the draft. The writer plans first (the reader's questions as headings) with the level's style guide and two example pairs written by the team; repair gets only the flagged blocks, their findings and facts. Answers are parsed leniently and validated with pydantic (one retry with the error); calls are streamed, retried on broken streams, 429 and 5xx, and logged.

Where Apertus fell short, and what does the job instead:

- The 8B drops must-keep facts while writing, ignored compound splits it was given, copied values from the prompt's example and broke JSON: code completes the fact sheet, splits compounds, returns answers with values the document lacks and repairs the JSON; facts rounds and the recover round ask again, a polish round repairs the sentences they add.
- The 70B judge's fact verdicts agree with blind labels hardly better than chance (κ 0.08, §5), so every fact number comes from code.
- On the local 4-bit build, Write with thinking took 24 minutes and paraphrased; without, 28 seconds and easy language. Write runs without thinking.
- 8B drafts read worse than the official versions (Amstad 51 against 62); the 70B writes cleaner (56) at twice the price per page.
- On a whole scanned page both models change digits (an IBAN into another IBAN); names and e-mail addresses are unsafe in every setting. Image input is not built (track_2b/docs/image-input.md).

Other models and coding assistants. No other model at runtime; spaCy's statistical parsers and wordfreq serve the checker. The 59 labels for the judge's agreement (§5) were made by Claude (Anthropic), blind to the verdicts; no pipeline score depends on them. Prompt tests without a key ran on a community 4-bit build of the 8B in llama.cpp (`Colby/apertus-v1.5-8b-text-Q4_K_M-GGUF`), never mixed with CSCS numbers. The code, tests and documents were written with AI coding assistants (Claude Code, Anthropic), directed and reviewed by the team; the assistants also ran the tests and the evaluation scripts; the design, the decisions and every claim here are the team's.

4. Data

Data	Origin	Licence and handling
Reference pairs: 35 documents with an official easy-language version (30 German: 20 `eval`, 10 `dev`; 5 French: 3 `eval`, 2 `dev`)	Cantons Zürich, Bern, St. Gallen, Luzern, Basel-Stadt, Graubünden, Zug, Geneva, Neuchâtel; federal offices BAG, BWL, EBGB	Personal use only. The repository holds only the manifest (`track_2b/data/pairs/manifest.jsonl`: URLs, licence notes); `eval fetch` caches the texts, git-ignored, with a fetch date
Example documents, few-shot examples, hard texts for tests	Written by the team: invented municipality, reserved `.example` domains	In the repository, under the project licence

`dev` pairs serve prompt development and calibration; `eval` pairs are held out. No personal data is used. Evaluating against protected pages needs no consent as long as neither the texts nor our rewrites are published (Copyright Act art. 24a and 24d; our reading, not legal advice; track_2b/docs/sources.md §4).

5. Evaluation

Task. Simplify a standard-language official text to Leichte Sprache and keep every fact. Each held-out pair runs under these conditions:

Code	Condition	Measures
P0	One prompt: “Schreiben Sie diesen Text in Leichter Sprache:”	What a user gets from a chatbot
P0-ZH	The Canton of Zurich’s public Leichte Sprache prompt, verbatim (`complete`: keep everything; `condensed`: essentials only, their default); German only	The strong single prompt: Zurich’s published prompt, the only public Swiss one we found
P1	Understand, Write, rule check; no judge, no repair	Fact sheet, planning, provenance
P2	Full pipeline with judge and repair	Checking and repairing

Two rows need no model: SOURCE (the original, scored as if it were a draft) and REF (the official easy-language version written by professionals).

Metrics, on every row’s final text: **fact coverage** (must-keep facts with a source value found); **hard facts kept** (the source’s countable values, by code, so comparable with REF); **silent losses** (hard values lost without a trace; for the pipeline, the gaps of its code completion); **values not in the source**; **unsupported blocks** (70B judge); **rule errors**; **readability** (Amstad, the German Flesch); **SARI** against the official version (a port of `SARI.py`, checked bit for bit); cost at Public AI’s token prices and speed per 250-word page.

Results on the 20 held-out German pairs, CSCS endpoint, final run of 11 October 2026 (commit 663b609), means over pairs; “/100” is per 100 blocks. The baselines ran with the cache off; P1 took P2’s Understand answers and first drafts from the cache, so its seconds are not measured, and the last row its Understand answers from the row above. The 70B everywhere ran on 19 pairs: the server cut its long Write call on one page in all 35 attempts (track_2b/eval/report.md has the full tables, also for French).

Setup	Fact coverage	Hard facts kept	Silent losses	Values not in source /100	Unsupported /100 (judge)	Rule errors /100	Amstad	SARI	USD / page	s / page
SOURCE	-	100 %	0 %	0.0	-	31.3	36	-	-	-
REF (official)	-	67 %	33 %	2.8	-	2.7	62	-	-	-
P0	92 %	93 %	7 %	0.2	-	19.5	52	43.0	0.0001	2.6
P0-ZH complete	83 %	85 %	15 %	0.0	-	15.0	47	43.0	0.0003	2.4
P0-ZH condensed	69 %	67 %	33 %	0.2	-	9.0	46	41.5	0.0002	1.4
P1	74 %	73 %	0 %	1.1	-	8.3	51	40.6	0.0014	-
P2 (ours)	**100 %**	**97 %**	**0 %**	**0.1**	**1.6**	**4.2**	**51**	**41.7**	**0.0131**	**63.3**
P2, 70B everywhere	99 %	92 %	0 %	0.0	1.1	0.6	56	43.8	0.0262	57.5
P2, 70B reads, 8B writes	100 %	95 %	0 %	0.1	1.0	1.9	53	42.0	0.0163	32.4

Reading the table. The one-sentence prompt and the strong prompt’s complete mode keep most values because they rewrite little (7.6 and 7.0 words per sentence; SARI rewards closeness to the source) and lose the rest without a trace; the condensed mode, its default, simplifies more and drops a third of the values. The pipeline nears the official sentence length (7.1 words against 6.1), keeps more hard facts than the official versions, which condense on purpose, and reads worse (51 against 62). Judge, repair and the recover round (P1 to P2) raise fact coverage from 74 to 100 percent and cut rule errors from 8.3 to 4.2 per 100 blocks; no must-keep fact is missing from the final versions (205 on the 23 pairs). Two fifths of P2’s rule errors come from one 7-page source, written out in full. On the 17

pairs with 3 or more hard facts P2 keeps 97 percent, the official versions 61. A single prompt varies from run to run: on unchanged templates, P0-ZH complete found 91 percent of the facts on 10 October and 83 on 11 October (one answer per pair at temperature 0.3). French (3 pairs): 100 percent fact coverage, 87 percent of the hard facts, 4.9 errors, readability 74 (official 85).

Cost, and ten times the volume. At Public AI's list prices the pipeline costs about one and a half US cents per 250-word page (P2: 0.0131 USD, the recover and polish rounds included; the 70B everywhere 0.0262), so ten times the volume costs ten times the cents. On own hardware the GPU is a fixed cost until it is busy. Speed is the limit: one job runs at a time, so at ten times the volume jobs wait in a queue; vLLM serves parallel requests, so more workers on the same GPU are a setting (`JOB_WORKERS`; not measured). The rule check takes 47 ms (median, a 15-block letter). The writer's time is the cost that matters: our own review of three P2 drafts in the UI (the two example letters and one dev pair, 1,079 source words together), from opening the review to the export, took 15 minutes by stopwatch, about 3.5 minutes per 250-word page (the team, not experts).

Judge reliability. 30 facts and 30 blocks from the P2 dev drafts of 6 October (70B judge), drawn evenly over the verdicts and labelled blind to them (`eval sample`, `eval kappa`). Blocks, supported or not: agreement 80 %, Cohen's κ 0.60. Facts, present, missing or distorted: 38 %, κ 0.08 (present or not: 62 %, κ 0.22). The fact verdicts are a hint for the reviewer, not a measure: every fact number here comes from code, and the review page shows a verdict that code contradicts as a doubt.

Checker calibration and tests. On official pages each error is a real deviation or a false positive to fix; calibration cut errors per 100 blocks from 12.6 to 2.0 (30 German pages) and from 5.3 to 2.2 (5 French). 1,375 offline tests include golden sentences for every rule (two or more that fire, one that must not).

6. Limitations

- **A2 with an 8B model.** Drafts near the official sentence length, not the readability (51 against 62), find 100 percent of the must-keep facts and keep 97 percent of the countable values (20 German eval pairs, CSCS), listing the rest.
- **Not a certification.** Easy-language texts must be tested by readers from the target group; the tool prepares that test, it does not replace it.
- **Italian** gets only the language-independent rules (facts, layout, readability). The B1 thresholds are our own; no official Swiss B1 rule set exists.
- **The judge is a model**, and obligations and conditions without a value rely on it alone; many pages carry few countable facts, so the fact-rich subset is reported too. On one GPU the 8B judges, which we have not evaluated.
- **Decision mode v1** is calibrated on one team-written KESB decision and one published court decision; no authority and no reader has seen its output.
- **Not yet supported:** scanned PDFs and photos (§3), DOCX export, an air-gapped image bundle, user accounts of its own (a shared token or the proxy's user header stand in).
- **Unmeasured:** the rewrite recipes the checker hands the writer for a subordinate clause, in German and French; the dev runs measured the rounds, not them.

7. Reproducibility

- **Run:** `LLM_BASE_URL`, `LLM_API_KEY` and `LLM_NAME` in the shell or in `.env`, then `make run` (http://localhost:8000); `make smoke` runs a bundled letter end to end.
- **Numbers:** `make eval` fetches the pairs, runs P0, P0-ZH, P1 and P2 with the response cache (an interrupted run resumes) and writes the tables; `make calibrate` runs the checker on the official pages without a model.
- **Environment:** Python 3.14 in `python:3.14-slim`, dependencies locked in `uv.lock`, spaCy models pinned by wheel URL; CI runs the linter, the tests and a start of the Docker image on every push.
- **Seeds:** temperatures above zero are not seeded; reruns reproduce exact outputs through the response cache, and each result stores the hash of its prompt template.
- **Hardware and commit:** the CSCS endpoint for all reported numbers except §2's L4 figures (`track_2b/eval/hardware/`); commit 663b609 (the final run of 11 October 2026, on the prompt templates with the document fence against prompt injection).

8. Next steps

- **Into use.** Before a canton's first real letter: its login proxy in front (the app reads the proxy's user header) and DOCX export (writers work in Word), to build; a GPU host and the weights under the acceptable use policy, to obtain; its IT security review and a data protection assessment, to approve. The target-group test stays with the office. First user: **[OWNER: none is named yet; say so plainly, or name one that agrees.]**
- **The same need elsewhere.** Every canton and federal office that publishes easy language, and texts under secrecy, where only Apertus fits (track_2b/docs/use-cases.md): decisions of KESB and courts (decision mode v1 exists; next an anonymisation step), patient letters, penal and payment orders, social, school and tax decisions. Most turn on a period, a reference number, an address or a dose, which code already checks like a date.
- Italian rules and prompts; image input in strips with a person checking every name and address; an air-gapped bundle; a LoRA on the 8B from approved drafts and edits.

License

Code: Apache License 2.0 (LICENSE). This report: Creative Commons Attribution 4.0 (CC-BY-4.0).

References

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